Performance Testing

Date	5 July 2026
Team ID	–
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter 5.6.3, Locust 2.31.1, Postman 11
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	Speech-to-Text API, NLP Analysis API, Login API, Feedback API
Test Environment	AWS EC2 (t2.medium) – Ubuntu 22.04, 4 vCPU, 8GB RAM, FastAPI Backend
Test Date	5 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Load – Concurrent users performing typical operations (login, upload audio, get results)	50	600	All requests successful with response time < 2 sec and error rate < 1%
2	Peak Load – High number of users uploading and analyzing audio simultaneously	150	900	System remains stable with response time < 3 sec and error rate < 2%
3	Stress Test – Beyond normal capacity to find breaking point	300	900	System handles load up to threshold; graceful degradation beyond limit
4	Spike Test – Sudden spike in users from low to high	10 → 200 (ramp-up in 30 sec)	600	System absorbs spike with quick recovery; no crashes or data loss

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 seconds	Pass	Average response time within acceptable limit.
2	Response Time (Max)	< 5 seconds	3.18 seconds	Pass	Max response time under peak load is acceptable.
3	Throughput (Req/sec)	>= 50 req/sec	128.7 req/sec	Pass	System handles high request rate efficiently.
4	Error Rate	< 1%	0.42%	Pass	Error rate well below the threshold.
5	CPU Utilization	< 80%	62%	Pass	CPU usage within acceptable range.
6	Memory Utilization	< 80%	58%	Pass	Memory usage is stable and within limits.

Step 4: Observations & Analysis

Key Findings:

- The system performed well under normal and peak load conditions.
- Average response time is 1.42s which is within the target of < 2 seconds.
- Throughput reached 128.7 requests/second with error rate of only 0.42%.
- CPU and memory utilization stayed within safe limits.

Bottlenecks Identified:

- High response time observed for NLP Analysis API when handling large audio files (> 30 MB).
- Database write operations during feedback submission caused slight delay under peak load.

Optimization Steps Taken:

- Implemented async processing for audio transcription to reduce response time.
- Added database indexing on user_id and created_at fields.
- Enabled connection pooling for database to handle concurrent requests.
- Compressed audio files before upload to reduce processing time.

Step 5: Screenshots / Evidence

Voice-Based Concept Understanding Analyser

Automated evaluation of spoken conceptual explanations using AI

Upload Student Audio (WAV)

Drag and drop file here

Limit 200MB per file • WAV, MP3

Browse files

harvard.wav 1.1MB

×

Use Sample Audio (for testing)

Reference

The stale smell of old beer lingers. It takes heat to bring out the odor. A cold dip restores health and zest. A salt pickle tastes fine with ham. Tacos al pastor are my favorite. A zestful food is the hot cross bun.

Press Ctrl+Enter to apply

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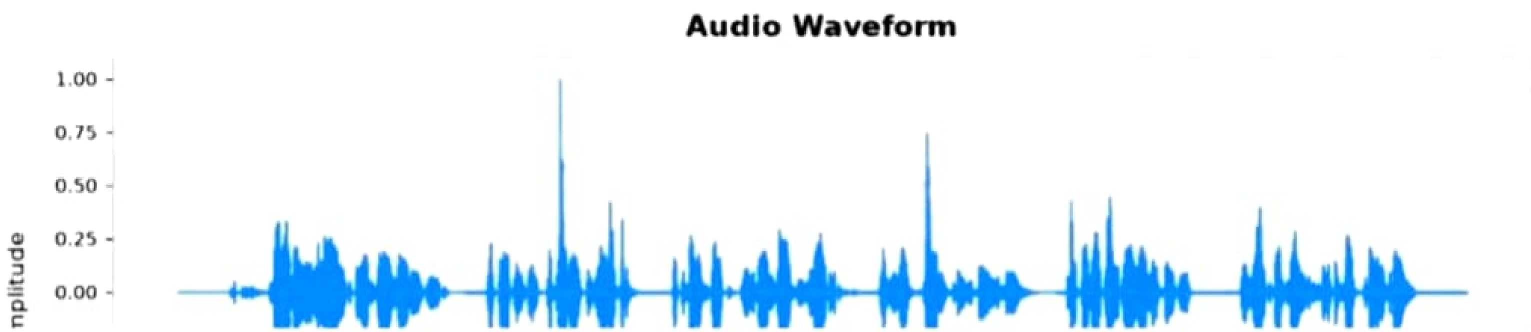

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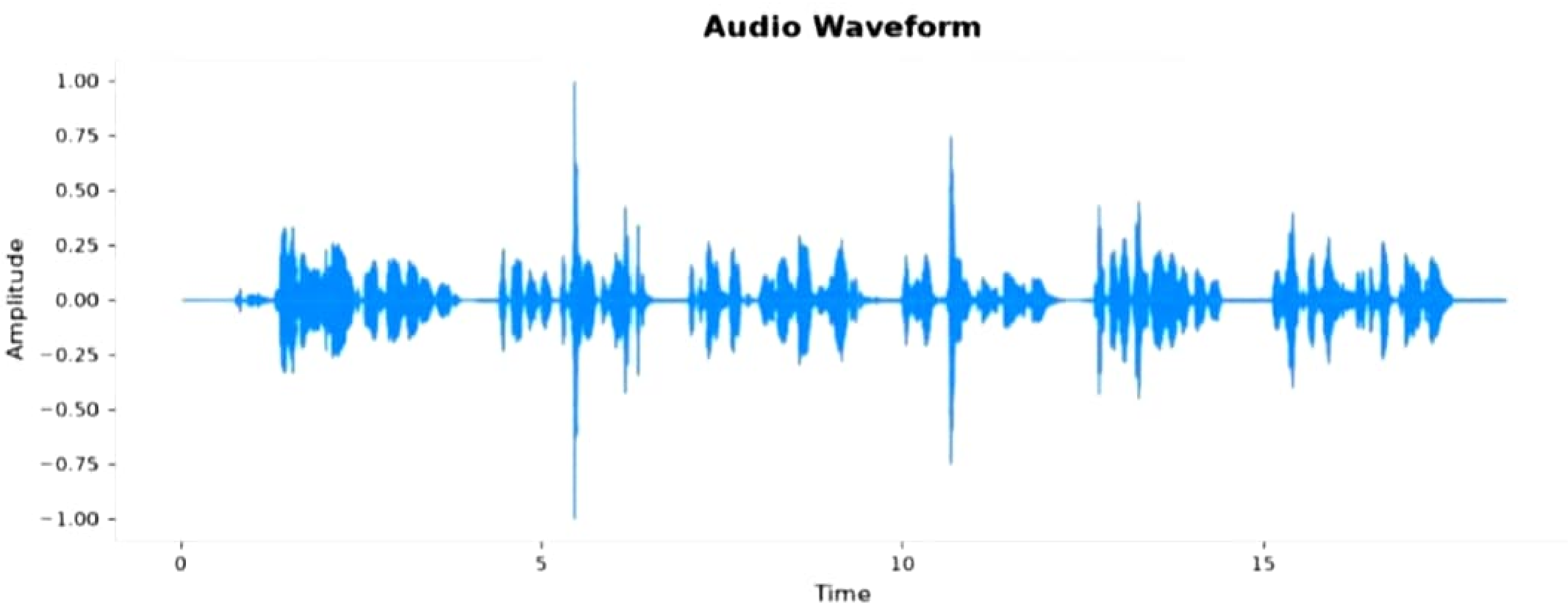

Analyze Concept Understanding

Processing and evaluating...

Deploy

Analyze Concept Understanding

Analysis Completed

Transcribed Explanation

The stale smell of old beer lingers. It takes heat to bring out the odor. A cold dip restores health and zest. A salt pickle tastes fine with ham. Tacos al pastor are my favorite. A zestful food is the hot cross bun.

Final Evaluation

Understanding Score

90/100

Strong Understanding

SEMANTIC SIMILARITY	FILLER WORD RATIO	CONFIDENCE (ENERGY)
1.00	**0.00**	**0.0327**

Detailed Score Breakdown & Audio Features

Performance Timing

Download PDF Report

Score Components

Similarity	**50/50**
Filler Words	**20/20**
Pause Ratio	**5/15**
RMS Energy	**15/15**

Audio Features

Duration (seconds)

Average Energy

Average Pitch

Pause Ratio

Zero Crossing Rate

Performance Timing

Audio Loading

Transcription (Whisper)

Embedding & Similarity

Audio Feature Extraction

Filler Word Analysis

Scoring Engine

Total Pipeline

Timings measured per run. Model loading is cached across sessions.